RES201 Week 1 Report
Chemical-Family Domain Shift in Pretrained ALIGNN

Prepared from workspace audit of `res201-alignn-domain-shift`

April 7, 2026

1 Purpose

This report documents the complete Week 1 execution status for the RES201 project *Chemical-Family Domain Shift in Pretrained ALIGNN Data Efficiency for Oxide vs. Nitride Formation Energy Prediction*. It is intended to serve two purposes:

1. provide a reproducible record of how the Week 1 deliverables were executed and verified; and
2. provide reusable methods text and artifact references for the final research paper.

The official Week 1 deliverables, taken from the project brief in `Project_Task/RES201 Project.pdf`, were:

- oxide subset filtered and confirmed;
- nitride subset filtered and confirmed with oxynitrides excluded from the nitride arm;
- original JARVIS split IDs applied before family filtering;
- fixed oxide and nitride test sets;
- pretrained checkpoint loaded;
- oxide zero-shot MAE reported;
- nitride zero-shot MAE reported;
- one oxide fine-tuning run at $N = 50$ completed;
- one nitride fine-tuning run at $N = 50$ completed.

2 High-Level Outcome

Week 1 is complete in this workspace. Both family datasets exist and validate successfully, both zero-shot evaluations exist, and both $N = 50$ fine-tuning runs exist with saved checkpoints, histories, test predictions, and summaries.

The most important scientific result from Week 1 is that under the current setup the $N = 50$ fine-tune runs performed worse than zero-shot for both families. This does **not** invalidate Week 1 completion; instead it becomes a key observation to investigate in Week 2 and later analysis.

3 Methodology

3.1 Stage 2 Dataset Construction

The dataset source used in this workspace is `dft_3d_2021` from JARVIS. The project rule from the brief was followed exactly:

1. recover or provide the original benchmark train/validation/test split IDs;
2. apply those split IDs first;
3. only then filter by chemical family.

The Stage 2 pipeline is implemented in:

- `scripts/dataset/build_res201_family_datasets.py`
- `scripts/dataset/res201_stage2_lib.py`
- `scripts/dataset/validate_res201_stage2.py`

The split manifest used for formation energy is:

- `manifests/dft_3d_formation_energy_peratom_splits.csv`

and the explicitly conflicting IDs are recorded in:

- `manifests/dft_3d_formation_energy_peratom_splits_conflicts.csv`

The family definitions implemented in the workspace are:

- oxide arm: any material containing O;
- nitride arm: materials containing N and not O;
- oxynitrides: excluded from the nitride arm;
- oxynitrides may still appear in the oxide arm because the brief defines oxides by O presence.

This choice is visible in the saved family summaries:

- oxide summary reports `oxynitride_count_in_family = 499`;
- nitride summary reports `oxynitride_count_in_family = 0`.

3.2 Stage 2 Validation

The validator was rerun directly on the current workspace:

```
python scripts/dataset/validate_res201_stage2.py --root ./data_shared
```

It returned `status = ok`. The validated family counts are:

Family	All	Train	Val	Test	Pool
Oxide	14991	11960	1547	1484	13507
Nitride	2288	1837	209	242	2046

These values match the saved summaries in `data_shared/*/summaries/summary.json`.

3.3 Stage 3 Environment and Pretrained Checkpoint

The Week 1 training environment is documented by:

- `env/bootstrap_res201_stage3_train.sh`
- `requirements/res201_train_frozen.txt`

The locally extracted pretrained formation-energy checkpoint used for Week 1 is:

- `jv_formation_energy_peratom_alignn/checkpoint_300.pt`
- `jv_formation_energy_peratom_alignn/config.json`

During the audit it was confirmed that the installed ALIGNN version in the `res201_train` environment differs in important ways from the assumptions in the earlier timeline notes and tutorial summary. Because of those API mismatches, the Week 1 fine-tuning path was repaired locally so that the current environment can reproducibly:

- load user-provided `id_prop.csv` plus POSCAR data from a local dataset root;
- load the local pretrained ALIGNN checkpoint correctly;
- freeze all parameters except the chosen final groups;
- train on the fixed train/validation/test partition for the chosen N.

3.4 Zero-Shot Evaluation Procedure

The zero-shot evaluation script is:

- `scripts/shared/zero_shot_alignn.py`

This script:

1. reads the fixed family test manifest;
2. loads each POSCAR structure from `data_shared/<family>/structures`;
3. runs `alignn.pretrained.get_prediction`;
4. coerces ALIGNN outputs to scalar floats;
5. computes test MAE in eV/atom;
6. writes both a predictions CSV and a summary JSON.

Representative command form:

```
source "$(conda␣info␣--base)/etc/profile.d/conda.sh"
conda activate res201_train
python scripts/shared/zero_shot_alignn.py --family oxide --repo-root .
python scripts/shared/zero_shot_alignn.py --family nitride --repo-root .
```

3.5 $N = 50$ Fine-Tuning Dataset Preparation

The Week 1 fine-tuning dataset builder is:

- `scripts/shared/prepare_week1_finetune_dataset.py`

This script constructs a local `dataset_root` for each family as follows:

1. read the fixed family `pool.csv` and `test.csv`;
2. sample $N = 50$ points from the pool using NumPy RNG with `seed = 0`;
3. allocate 45 samples to train and 5 samples to validation;
4. keep the family test set fixed and unchanged;
5. copy the corresponding POSCAR files into `results/<family>/N50_seed0/dataset_root`;
6. write `id_prop.csv` and `split_manifest.json`.

Representative commands used:

```
python scripts/shared/prepare_week1_finetune_dataset.py \
  --family oxide --N 50 --seed 0 --repo-root . --link-mode copy

python scripts/shared/prepare_week1_finetune_dataset.py \
  --family nitride --N 50 --seed 0 --repo-root . --link-mode copy
```

3.6 $N = 50$ Fine-Tuning Configuration

The config writer is:

- `scripts/shared/write_week1_alignn_config.py`

It writes a TrainingConfig-compatible JSON for the currently installed ALIGNN version with:

- `dataset = user_data`;
- `target = target`;
- `filename = id_prop.csv`;
- `epochs = 50`;
- `batch_size = 16`;
- `learning_rate = 1e-4`;
- fixed `n_train`, `n_val`, and `n_test`;
- the pretrained ALIGNN model architecture copied from the local checkpoint config.

The generated Week 1 configs are:

- `configs/oxide_week1_N50_seed0.finetune_last2.json`
- `configs/nitride_week1_N50_seed0.finetune_last2.json`

3.7 $N = 50$ Fine-Tuning Training Procedure

The training wrapper used for Week 1 is:

- `scripts/shared/finetune_last2_alignn.py`

This wrapper was used because the installed ALIGNN CLI and lower-level APIs did not directly support the desired pretrained user-data fine-tuning path for this checkpoint version. The Week 1 wrapper performs the following:

1. load the local pretrained checkpoint from `checkpoint_300.pt`;
2. rebuild the ALIGNN model from the saved checkpoint config;
3. freeze all model parameters;
4. unfreeze only the final two parameter groups used in this workspace: `gcn_layers.3` and `fc`;
5. build LMDB-backed train/validation/test graph datasets from the local `dataset_root`;
6. train for 50 epochs using AdamW and one-cycle LR scheduling;
7. save checkpoint, history, and test prediction artifacts.

Representative commands used:

```
python scripts/shared/finetune_last2_alignn.py \
  --config configs/oxide_week1_N50_seed0.finetune_last2.json \
  --output-dir results/oxide/N50_seed0/finetune_last2 \
  --dataset-root results/oxide/N50_seed0/dataset_root \
  --pretrained-checkpoint jv_formation_energy_peratom_alignn/checkpoint_300.pt \
  --pretrained-config jv_formation_energy_peratom_alignn/config.json \
  --device cpu

python scripts/shared/finetune_last2_alignn.py \
  --config configs/nitride_week1_N50_seed0.finetune_last2.json \
  --output-dir results/nitride/N50_seed0/finetune_last2 \
  --dataset-root results/nitride/N50_seed0/dataset_root \
  --pretrained-checkpoint jv_formation_energy_peratom_alignn/checkpoint_300.pt \
  --pretrained-config jv_formation_energy_peratom_alignn/config.json \
  --device cpu
```

4 Week 1 Results

4.1 Zero-Shot Results

Family	Test size	Zero-shot MAE (eV/atom)
Oxide	1484	0.0341836068
Nitride	242	0.0695420150

4.2 $N = 50$ Fine-Tuning Results

Family	Train	Val	Test	Test MAE (eV/atom)
Oxide	45	5	1484	0.0700507342
Nitride	45	5	242	0.1617631432

4.3 Interpretation

For this first Week 1 pass, both zero-shot baselines outperform the corresponding $N = 50$ fine-tuning runs. That means Week 1 succeeded as an experimental milestone, but the numerical outcome suggests that:

- the present fine-tuning protocol is not yet data-efficient at $N = 50$;
- additional seeds, larger N, and from-scratch baselines remain essential;
- the current result should be reported honestly rather than hidden.

5 Week 1 Deliverable Checklist

Deliverable	Status	Evidence
Oxide subset filtered and confirmed	Done	`data_shared/oxide/manifests/*.csv`, `data_shared/oxide/summaries/summary.json`
Nitride subset filtered and confirmed	Done	`data_shared/nitride/manifests/*.csv`, `data_shared/nitride/summaries/summary.json`
Oxynitrides excluded from nitride arm	Done	nitride summary reports `oxynitride_count_in_family = 0`
Original split IDs applied first	Done	split source in summaries points to `manifests/dft_3d_formation_energy_peratom_splits.csv`
Fixed oxide test set	Done	`data_shared/oxide/manifests/test.csv`
Fixed nitride test set	Done	`data_shared/nitride/manifests/test.csv`
Pretrained checkpoint loaded	Done	`jv_formation_energy_peratom_alignn/checkpoint_300.pt`
Oxide zero-shot MAE reported	Done	`results/oxide/zero_shot/summary.json`
Nitride zero-shot MAE reported	Done	`results/nitride/zero_shot/summary.json`
One oxide fine-tune at $N = 50$ completed	Done	`results/oxide/N50_seed0/finetune_last2/summary.json`
One nitride fine-tune at $N = 50$ completed	Done	`results/nitride/N50_seed0/finetune_last2/summary.json`

6 Verification Commands

The simplest verification command is now:

```
scripts/shared/check_week1_status.sh .
```

Additional direct checks:

```
python scripts/dataset/validate_res201_stage2.py --root ./data_shared

python -m json.tool results/oxide/zero_shot/summary.json
python -m json.tool results/nitride/zero_shot/summary.json
python -m json.tool results/oxide/N50_seed0/finetune_last2/summary.json
python -m json.tool results/nitride/N50_seed0/finetune_last2/summary.json

python - <<'PY'
```

```
import json
for p in [
    "results/oxide/N50_seed0/finetune_last2/history_val.json",
    "results/nitride/N50_seed0/finetune_last2/history_val.json",
]:
    h = json.load(open(p))
    print(p, len(h), h[-1])
PY
```

7 Workspace Inventory

7.1 Top-Level Files and Folders

Path	Description
`.codex`	Local Codex marker file in this workspace.
`.git`	Git metadata for the local repository clone.
`.gitignore`	Ignore rules controlling which generated artifacts should stay local.
`Project_Task/`	Course and project reference PDFs: project brief, ALIGNN tutorial, and timeline notes.
`README.md`	Original Stage 2 payload README.
`cache/`	Local JARVIS download/cache area.
`configs/`	Saved Week 1 configuration JSON files, including historical failed attempts and final working fine-tune configs.
`data_shared/`	Canonical Stage 2 family datasets, manifests, summaries, diagnostics, structures, and ALIGNN-ready exports.
`docs/`	Stage 2 and Stage 3 playbooks plus frozen dataset notes.
`env/`	Environment bootstrap and validation scripts for the data and training environments.
`id_prop.csvtrain_data/`, `id_prop.csvval_data/`, `id_prop.csvtest_data/`	Root-level LMDB caches created by ALIGNN training utilities. These are local/generated artifacts, not canonical research outputs.
`jv_formation_energy_peratom_alignn/`	Local pretrained checkpoint directory containing the extracted formation-energy checkpoint and its config.
`manifests/`	Official split manifest, conflict list, and split manifest templates.
`notebooks/`	Currently empty notebook area reserved for later analysis.
`reports/`	Reporting area. This report and generated inventory files belong here.
`requirements/`	Frozen dependency record for the training environment.
`res201_stage2_data_shared.zip`	Local archive copy of the Stage 2 shared dataset payload.
`results/`	Week 1 outputs, including zero-shot predictions and $N = 50$ oxide/nitride experiment folders.
`scripts/`	Dataset pipeline scripts and shared Week 1 experiment scripts.

7.2 Key Directory Contents

Project_Task

- `RES201 Project.pdf`: official project brief and weekly deliverables.
- `ALIGNN_Tutorial.pdf`: usage-oriented tutorial/reference for ALIGNN.
- `Project_Timeline_So_Far.pdf`: human-written status summary from before this audit.

configs

- `oxide_week1_N50_seed0.json`: earlier experimental config draft.
- `oxide_week1_N50_seed0.fixed.json`: schema-compatible config produced during earlier debugging.
- `oxide_week1_N50_seed0.run.json`: another historical run config.
- `oxide_week1_N50_seed0.finetune_last2.json`: final working oxide Week 1 config.
- `nitride_week1_N50_seed0.finetune_last2.json`: final working nitride Week 1 config.

docs

- `STAGE2_PLAYBOOK.md`: Stage 2 instructions.
- `STAGE2_FROZEN_DATASET.md`: frozen dataset note file.
- `STAGE3_PLAYBOOK.md`: Week 1 training output expectations.

env

- `bootstrap_res201_stage2_data.sh`: create the Stage 2 data environment.
- `verify_res201_stage2_data.py`: verify Stage 2 environment readiness.
- `bootstrap_res201_stage3_train.sh`: create the Stage 3 training environment.

manifests

- `dft_3d_formation_energy_peratom_splits.csv`: official split mapping used for this project.
- `dft_3d_formation_energy_peratom_splits_conflicts.csv`: ambiguous IDs excluded conservatively.
- `original_split_manifest_template.csv`: template for user-supplied split manifests.
- `original_split_manifest_template.json`: JSON split template.

scripts/dataset

- `build_res201_family_datasets.py`: main Stage 2 dataset builder.
- `make_split_manifest_from_benchmark.py`: helper for constructing split manifests.
- `materialize_alignn_root.py`: materialize ALIGNN-ready directories.
- `res201_stage2_lib.py`: shared Stage 2 library logic.
- `validate_res201_stage2.py`: Stage 2 validator.

scripts/shared

- `zero_shot_alignn.py`: zero-shot evaluator.
- `prepare_week1_finetune_dataset.py`: construct local $N = 50$ experiment roots.
- `write_week1_alignn_config.py`: write TrainingConfig-compatible JSONs.
- `finetune_last2_alignn.py`: repaired fine-tuning wrapper for this ALIGNN environment.
- `save_pretrained_state_dict.py`: helper for extracting pretrained model weights.
- `check_week1_status.sh`: end-to-end Week 1 verification script.
- `generate_workspace_inventory.py`: generate the machine-readable full workspace inventory.

data_shared

- `diagnostics/`: global split manifest, schema report, and catalog diagnostics.
- `oxide/manifests/`: `all.csv`, `train.csv`, `val.csv`, `test.csv`, `pool.csv`.
- `oxide/summaries/summary.json`: saved oxide Stage 2 summary.
- `oxide/structures/`: one POSCAR file per oxide-family record.
- `oxide/alignn_ready/`: pool and test exports for ALIGNN consumption.
- `nitride/manifests/`: `all.csv`, `train.csv`, `val.csv`, `test.csv`, `pool.csv`.
- `nitride/summaries/summary.json`: saved nitride Stage 2 summary.
- `nitride/structures/`: one POSCAR file per nitride-family record.
- `nitride/alignn_ready/`: pool and test exports for ALIGNN consumption.

results

- `results/oxide/zero_shot/`: oxide predictions CSV and summary JSON.
- `results/nitride/zero_shot/`: nitride predictions CSV and summary JSON.
- `results/oxide/N50_seed0/dataset_root/`: local oxide Week 1 POSCAR subset, `id_prop.csv`, and split manifest.
- `results/nitride/N50_seed0/dataset_root/`: local nitride Week 1 POSCAR subset, `id_prop.csv`, and split manifest.
- `results/oxide/N50_seed0/finetune_last2/`: completed oxide Week 1 fine-tune outputs.
- `results/nitride/N50_seed0/finetune_last2/`: completed nitride Week 1 fine-tune outputs.
- `results/oxide/N50_seed0/train_alignn_restart/`: older partial attempt from the original debugging path.

7.3 Full Inventory File

Because the workspace contains thousands of generated POSCAR files and many derived artifacts, a full machine-generated inventory should accompany this report rather than being expanded inline. The generated inventory artifacts for this audit are:

- `reports/workspace_inventory.csv`
- `reports/workspace_inventory_summary.json`

They record every non-`.git` path in the workspace. At the time of generation, the inventory contained:

- `19,298` total paths;
- `19,243` files; and
- `55` directories.

The CSV provides one row per path with a path string, item kind, and short description so that the full workspace can be audited without manually expanding thousands of structure files in the PDF itself.

8 Git and GitHub Recommendation

The repository remote is:

- https://github.com/TheArchitect999/res201-alignn-domain-shift

Yes, the repository should be updated, but **not by pushing everything blindly**. The correct policy is:

- **Push:** scripts, configs, manifests, docs, summaries, lightweight prediction CSVs, report sources, and verification scripts.
- **Do not push:** heavy structure dumps, LMDB caches, model checkpoints, cached downloads, zipped local payloads, and large generated POSCAR collections inside experiment roots.

To make this safer, the local `.gitignore` was tightened during this audit to exclude:

- root-level LMDB caches;
- pretrained checkpoint binary;
- heavy result checkpoints and LMDB caches;
- local Stage 2 zip archive;
- copied POSCARs inside `results/*/dataset_root`.

Recommended push workflow:

```
git status
git add .gitignore
git add scripts/shared/check_week1_status.sh
git add scripts/shared/finetune_last2_alignn.py
git add scripts/shared/write_week1_alignn_config.py
git add configs/oxide_week1_N50_seed0.finetune_last2.json
git add configs/nitride_week1_N50_seed0.finetune_last2.json
git add results/oxide/zero_shot/summary.json
git add results/oxide/zero_shot/predictions.csv
git add results/nitride/zero_shot/summary.json
git add results/nitride/zero_shot/predictions.csv
git add results/oxide/N50_seed0/finetune_last2/*.json
git add results/oxide/N50_seed0/finetune_last2/*.csv
git add results/nitride/N50_seed0/finetune_last2/*.json
git add results/nitride/N50_seed0/finetune_last2/*.csv
git add reports/week1_report.tex
git add reports/workspace_inventory_summary.json
git add reports/workspace_inventory.csv
git commit -m "Complete␣RES201␣week␣1␣oxide␣and␣nitride␣experiments"
git push origin main
```

Before pushing, run `git status` and verify that no huge `.pt`, `.vasp`, LMDB cache, or archive files are staged.

9 Conclusion

The workspace now satisfies the Week 1 requirements for both the oxide and nitride arms. The work completed in this repository is sufficient to claim Week 1 completion, provided the team reports the outcomes honestly:

- Stage 2 family datasets are valid and frozen;
- zero-shot results are recorded for both families;
- one $N = 50$ fine-tuning run has been completed for each family;
- all required evidence files exist and can be verified from the terminal.

The next scientific priority is not Week 1 completion, which is already done, but explaining why the current $N = 50$ fine-tunes underperform zero-shot and extending the study across additional N values and seeds.